Rising biocapacity total can mask ecological-capital land drawdown: a two-land model

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Abstract

Biocapacity is reported as an aggregate, despite wide differences in yield and regeneration among its constituent land classes. A rising total alone cannot reveal whether its composition has shifted. We examine this accounting problem with a dynamic two-land model that includes delayed ecological regeneration and population response. Under the model's baseline conditions, a shortfall-triggered transfer to fast-provisioning land with a higher annual contribution per hectare adds to model biocapacity through the conversion term even as capital-land area falls. In a representative run, total model biocapacity rises 39% as capital-land area falls 46.16%; the modeled capital-land contribution falls 51.72%. On the reduced fixed-demand branch, the balance relation reaches a minimum-area floor without a fold, hysteresis, or critical slowing; this result does not characterize every trajectory of the delayed model. Separately, NFA Edition 2026 account values for 1961–2023 show total world biocapacity up 24.86%, while B/P falls 52.71% and the production footprint per person rises 16.73%. Cropland contributes +24.06 percentage points, Forest biocapacity −5.95, and built-up land +7.44 as an accounting class. These account movements are not measurements of physical stocks. Component values help interpret the aggregate.

Keywords: biocapacity, ecological footprint, land-use conversion, aggregation, identifiability, critical slowing down, delay differential equations, carrying capacity

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1. Introduction

In ecology, *carrying capacity* describes the population a resource base can sustain. Many models treat it as a fixed ceiling, in part because it is difficult to estimate (Cohen, 1995). If resource use changes the basis of future production, however, a fixed ceiling may not capture that change. This raises a measurement question: can a headline environmental metric report a rising total while a productive land class loses area?

The National Footprint and Biocapacity Accounts (NFA) report biological supply as biocapacity and human demand as the Ecological Footprint. Both are expressed in global hectares (gha), which adjust area for biological productivity relative to the world average in a given year; the annual basis is implicit (Global Footprint Network, 2009, 2019). Because a global hectare combines area and productivity, a change in the total does not by itself show whether yields, area, or the mix of land uses changed. Has productivity increased, has the area in production shifted, or has the composition of the account changed?

In composite-indicator theory, this ambiguity is called *compensation*: a gain in one component can offset a loss in another (Nardo et al., 2008; Becker et al., 2017; Fischer et al., 2022). The aggregate remains useful, but reading it alone can conceal component changes. Related critiques discuss the interpretation of the Ecological Footprint and biocapacity accounts (Blomqvist et al., 2013; Giampietro and Saltelli, 2014; van den Bergh and Grazi, 2015; Galli et al., 2016).

Most human–environment models summarize nature as an aggregate stock or a unified feedback. Examples include the *World3* simulations (Meadows et al., 1972, 1974, 2004), models of Easter Island (Brander and Taylor, 1998; Anderies, 2000), and coupled Earth-system frameworks (Motesharrei et al., 2016; Donges et al., 2017; Calvin and Bond-Lamberty, 2018; Henderson and Loreau, 2018). Other work debates whether human carrying capacity is fixed, variable, or useful as a concept (Cohen, 1995; van den Bergh and Rietveld, 2004; Simon, 1996; Meyer, 1994; Meyer and Ausubel, 1999; Dasgupta, 2021); global demographic assessments often project population separately from environmental constraints (United Nations, 2022). Land-use studies, meanwhile, document conversion among forests, grasslands, wetlands, and cropland (Haberl et al., 2007; Krausmann et al., 2013; Erb et al., 2017). Together, these literatures motivate a narrower question: what can one biocapacity total reveal when land classes differ in yield and regeneration?

The orchard and the elevator. To make the accounting issue concrete, imagine an orchard with fast-provisioning ground, such as intensively managed beds, beside slower-renewing ground with trees and living soil. Annual harvest is a *flow*; trees, roots, and soil are assets that support future flows. In our model, a shortfall can trigger a transfer from ecological-capital land to fast-provisioning land. If the receiving class has a higher annual contribution per hectare in that scenario, the conversion term adds to B at the margin while capital-land area A_c falls; other flows can offset that increment. This is a conditional model mechanism, not evidence that NFA accounts measure physical stocks or that a particular real-world conversion occurred. Replanting may restore a later harvest, but it is not a one-for-one measure of the condition or recovery of the distinct assets represented by the model's capital-land class. The analogy separates current output from the state of its productive base; it does not assert that the classes map directly to NFA categories.

A second analogy separates observed service from structural condition: an elevator rated for ten people might continue operating while carrying fourteen, but the ride itself does not establish whether repeated overloading has changed the cable's safety margin. Continued service is not a measurement of structural condition, nor does the analogy predict inevitable failure. Related material-ledger work discusses a broader flow–stock masking problem; here we study its composition-specific form in a biocapacity model (Abaee, 2026c).[1]

Research questions. This paper addresses three questions:

1. **The masking dynamic:** Under what conditions can conversion raise total model biocapacity while A_c declines, how long can this divergence last, and can the aggregate identify the land shift?

2. **System trajectory:** Under the model's stated assumptions, does drawdown involve a fold, hysteresis, or critical slowing, or does the reduced comparison follow another path?

3. **Limits and recovery:** What limits drawdown under fixed demand, and what modelled process allows capital-land area to recover?

[1]The v50 PDF is present at the pinned commit. The checksum manifest lists the v49 article files but omits v50; the README dated 2026-09-19 does not explicitly label v49 or v50; and the v50 TeX header points to the v48 Markdown source.

Core contributions. We make three contributions:

- We construct a dynamic two-land model that conserves total area, represents capital-land quality separately from area, and includes delays in ecological regeneration and demographic response.
- We derive the condition under which a conversion term raises aggregate biocapacity while A_c falls. In a reduced fixed-demand comparison, the balance relation reaches the area floor without a fold or critical slowing; realized paths in the delayed model remain history-dependent. These are conditional model results, not general claims about observed ecosystems.
- We decompose NFA Edition 2026 account values for 1961–2023 and examine a separate cropland-area/yield proxy. The account series show a compositional change, not a census of physical ecological stocks.

Finally, we derive five testable empirical predictions. The model parameters are representative rather than fitted, so simulations illustrate conditional mechanisms rather than historical forecasts. Aggregate biocapacity is therefore most informative when read alongside its component accounts.

2. Methods

2.1. A two-land model with conserved area and ecosystem quality

To examine how an aggregate indicator can mask a change in composition, we divide total bioproductive area A_{tot} among three mutually exclusive classes:

- **Fast-provisioning land** (A_f): cropland and other intensively managed systems, represented by their annual contribution.
- **Ecological-capital land** (A_c): forest, soil, wetland, and long-rotation systems grouped into one effective class, rather than treated as a single real land type.
- **Reserve land** (A_r): fallow, retired, or not-yet-allocated area.

The three areas sum to A_{tot}. The model also assigns capital land a separate normalized quality state $q \in [0, q_{\max}]$. This effective state changes the

modeled annual contribution per hectare; it is not a direct measurement of biomass density, soil integrity, or ecological health. Quality growth follows a logistic rule:

$$G_c(q) = \rho_c \, q \left(1 - \frac{q}{q_{\max}}\right), \tag{1}$$

Capital land's annual contribution per hectare combines a direct-yield term with a term linked to quality growth:

$$\Upsilon_c(q) = b_c + b_{G,c} \, G_c(q), \tag{2}$$

where b_c is direct yield and $b_{G,c}$ converts the quality-growth term into annual account value. The fast- and capital-land contributions, Y_f and Y_c, sum to total model biocapacity:

$$B = Y_f + Y_c = b_f A_f + \Upsilon_c(q) \, A_c, \tag{3}$$

Here b_f is fast-land contribution per hectare. Model B is an annual account-equivalent amount in gha, with the annual basis implicit (Global Footprint Network, 2009, 2019).

Demand, shortfalls, and land conversion. Total demand is $E = eP$, where P is population and e is the constant footprint per person (gha per person; cap denotes one person in the equations and tables). The shortfall S is the demand not met by the human-available shares of the two land contributions:

$$S = \left[E - \sigma_f Y_f - \sigma_c Y_c\right]_+, \tag{4}$$

The positive-part operator is $[x]_+ = \max(x, 0)$; baseline availability fractions are $\sigma_f = \sigma_c = 1$. When $S > 0$, the model transfers capital land to fast-provisioning land at a finite rate:

$$u_c = \frac{1}{\tau_{\text{conv}}} \min\left(\frac{S}{b_f}, \; A_c - A_c^{\min}\right) \qquad [\text{ha yr}^{-1}], \tag{5}$$

The ratio S/b_f is the fast-land area required to cover the shortfall under this model rule; $A_c^{\min}$ is the lower area bound, and τ_{conv} sets the conversion timescale. This transfer is a specified response rule, not a reconstruction of observed land-clearing decisions.

Land flows and asymmetric restoration. Land moves among classes through one-to-one area transfers:

$$\dot{A}_c = -u_c + R_{rc}, \tag{6}$$

$$\dot{A}_f = u_c + \mu A_r - \eta_f A_f, \tag{7}$$

$$\dot{A}_r = -\mu A_r + \eta_f A_f - R_{rc}, \tag{8}$$

The flow μA_r moves reserve area into fast use, while $\eta_f A_f$ retires fast land to reserve. Reserve land returns to the capital-land class only through the surplus-gated restoration flow R_{rc}:

$$R_{rc} = \chi_r \, A_r \, [B - E]_+ . \tag{9}$$

Restoration is therefore asymmetric in the model: fast land may be retired into reserve, but reserve returns to ecological-capital use only when the modeled surplus is positive and the restoration flow is active. A closed shortfall alone does not restore capital-land area.

Delays, model debt, and demographics. Quality responds to its delayed value, and population follows a delayed logistic law (Pearl, 1925; Hutchinson, 1948) with carrying capacity $K = \max(B/e, K_{\min})$. The model also tracks a leaky deficit state D that affects fast-land yield:

$$\dot{q} = G_c\big(q(t - \tau_g)\big), \tag{10}$$

$$\dot{P} = rP\left(1 - \frac{P}{K(t - \tau_p)}\right), \qquad K = \max(B/e,\, K_{\min}), \tag{11}$$

$$b_f = \big(b_{f0} + T_b(t)\big)\, e^{-\alpha D}, \tag{12}$$

$$\dot{D} = [E - B]_+ - \eta D, \tag{13}$$

$$T_b(t) = \omega_{Tb}\, \frac{\Delta b}{1 + e^{-\kappa_w (t - t_{wave})}}, \qquad \omega_{Tb} \in \{0, 1\}. \tag{14}$$

Areas A_f, A_c, and A_r are measured in hectares; B, E, and S are annual account amounts in gha, with the annual basis implicit. D is measured in gha yr; both $\dot{D}$ and ηD are in gha. It is an internal model state driven by aggregate deficit, not an area, a physical stock, or an NFA-reported quantity. The optional technology boost T_b raises fast-land yield along a bounded logistic curve. It is off in the baseline, figures, and tables, and appears only in the matched comparison in SI §S8. The dynamic states are A_f, A_c, q, P, and D; timber liquidation and direct restoration from fast land to capital land are not active in the reported runs (SI §S1).

Core system feedbacks. The equations permit two conditional feedbacks:

1. **Compositional-yield feedback.** When fast land contributes more per hectare than capital land ($b_f > \Upsilon_c(q)$), conversion raises B at the margin while reducing A_c. With endogenous population, higher B can raise delayed carrying capacity K and later population and demand, potentially creating another shortfall. Maintenance, retirement, quality change, and the other model terms also affect B; this is a model feedback, not an empirical causal finding.

2. **Deficit–yield feedback.** A shortfall adds to D, while repayment draws it down. As D grows, modeled fast-land yield b_f falls; this can reduce B and deepen the shortfall. The state D is the model's leaky deficit measure, not a direct measure of nutrient depletion, aquifer overdraft, or other physical processes.

2.2. Parameter selection and empirical grounding

Table 1 summarizes the baseline parameter values and their evidence status. Parameters are representative rather than econometrically fitted; the model is designed to isolate structural mechanisms, not to generate precise historical forecasts. To keep the model tractable, it consolidates forests, soils, wetlands, and other slow-renewing systems into one effective capital-land class, although their extent, condition, yields, and recovery times differ substantially. Empirical records provide timescale context: secondary forest biomass reaches half its reference level after a median of about 30 yr; trajectories for 33 collapsed fish stocks give a median time to half-recovery of about 5.6 yr; and soil organic carbon approaches a new balance over roughly 17–23 yr (SI §S5.5) (Poorter et al., 2016; Hutchings and Reynolds, 2004; Neubauer et al., 2013; Poeplau et al., 2011). These heterogeneous records do not define a pooled global average, estimate ρ_c, or determine the model delay τ_g.

Two data-based quantities provide context for cropland but do not calibrate the model: the NFA cropland account value per unit area and a separate cereal-yield/cropland-area proxy (SI §S5). Their boundaries do not map one-to-one onto model fast land, and neither is a fitted value of b_f or a direct measure of ecological capital. Finally, the baseline assumes $b_f - \Upsilon_c(q) > 0$. This is an explicit conditional modeling regime for examining when a short-term contribution premium can mask a change in land composition, not a universal empirical claim.

Table 1: Baseline parameter values and evidence status. Representative values are effective, lumped quantities; the NFA account/area quotient is descriptive, not a physical-yield measurement. Extension parameters are active only in scenarios specified in this table or the SI. Symbols and units are listed in SI §S1.

Parameter	Value	Meaning	Status
ρ_c	$0.08\,\mathrm{yr}^{-1}$	quality regeneration rate of capital land	representative; not fitted
b_{f0}	$0.85\,\mathrm{gha\,ha}^{-1}$	base fast-land yield per hectare	representative (2026 NFA account value/area quotient ≈ 1.045 in 1961; not a physical-yield measure)
b_c	$0.05\,\mathrm{gha\,ha}^{-1}$	capital-land direct yield	representative
$b_{G,c}$	$0.8\,\mathrm{gha\,yr\,ha}^{-1}$	regeneration-value coefficient for capital land	representative
$q_{\max}$	1.0 (dimensionless)	normalized quality ceiling	normalization
μ	$0.06\,\mathrm{yr}^{-1}$	cropland maintenance (reserve→fast)	representative
η_f	$0.05\,\mathrm{yr}^{-1}$	fast-land retirement (fast→reserve)	representative
τ_{conv}	1.0 yr	conversion time constant	representative
α	$0.03\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$	degradation of b_f per unit debt	representative
η	$0.05\,\mathrm{yr}^{-1}$	debt-repayment rate (> 0)	representative
e	$0.55\,\mathrm{gha\,cap}^{-1}$	per-capita footprint (constant)	representative
r	$0.02\,\mathrm{yr}^{-1}$	intrinsic population growth	representative
$\tau_g,\ \tau_p$	20, 25 yr	regeneration and demographic delays	representative; τ_p lumps institutional lags
A_{tot}	2.5 ha	total bioproductive area	normalization
$\sigma_f,\ \sigma_c$	1, 1 (dimensionless)	human-available shares of annual account amounts	baseline (< 1: reservation extension)
χ_r	$0.10\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$	surplus-gated restoration coefficient	on with endogenous demand; off under frozen demand
$A_c^{\min}$, $A_r^{\min}$, $K_{\min}$	0.05 ha, 0.10 ha, 0.02 cap	typed floors	domain constraints
Δb, κ_w, t_{wave}	$1.0\,\mathrm{gha\,ha}^{-1}$, $0.1\,\mathrm{yr}^{-1}$, 40 yr	technology wave	off in baseline; active only in the matched SI S8 experiment

2.3. Data and empirical measures

Our empirical analysis uses the National Ecological Footprint and Biocapacity Accounts (NFA), Edition 2026, results and metadata version 1.0 (Lo et al., 2026). We analyse complete world observations for 1961–2023 and exclude preliminary estimates for 2024–2025. A 1961–2022 sensitivity check uses the same account edition and population series, so the data vintage is held constant across the two windows.

From the workbook's `world_data` sheet, we extract total biocapacity B (`biocap_total_gha`), the Global Ecological Footprint of Production E_{prod} (`efp_total_gha`), and UN PopStat population P (`pop_1000_persons`). Population is converted from thousands to persons. Per-person measures are calculated from unrounded totals rather than the workbook's rounded per-person columns; ratios, indices, and log changes are dimensionless.

Accounting categories and physical stocks. We retain the five workbook labels: *Cropland*, *Grazing land*, *Forest biocapacity*, *Fishing grounds*, and *Built-up land*. These are account values, not measurements of biomass, soil condition, wetland extent, fish biomass, or model area A_c. Built-up land is an NFA accounting class, not ecological capital. We therefore interpret movements among categories as changes in account composition, not as direct observations of physical ecosystems or of the model's capital-land stock.

Auxiliary cropland proxy. As a separate comparison, we pair Our World in Data's FAOSTAT-derived world cropland-area series (arable land plus permanent crops) with cereal yield (Our World in Data, 2026a,b). Cereal yield is a limited productivity indicator, not a measure of all-crop output or an NFA yield factor. We scale the proxy to the NFA cropland share of total biocapacity in 1961, calculate the crop-account proxy $X_t = \hat{b}_{f,t} A_{f,t}$, and define the remainder as $C_t = B_t - X_t$. Here $A_{f,t}$ denotes observed cropland area for this auxiliary calculation, not the model state. The scaling makes this an assumed account split, not an independent measurement of the model's fast-land contribution or of physical capital. The remainder absorbs differences in boundaries, categories, and calibration; it is not $\Upsilon_c(q)A_c$. SI §S5 reports the proxy and its calibration sensitivity. We infer no physical-stock trend from it.

Total and per-person change. We use the exact identity

$$\Delta \ln B = \Delta \ln(B/P) + \Delta \ln P$$

to separate the scale change in total biocapacity from the change in biocapacity per person. This is an accounting identity, not a causal model of biocapacity production.

2.4. Numerical implementation

We integrate the delay equations using the method of steps with an explicit first-order Euler scheme. Baseline trajectories and threshold bisections use $\Delta t = 0.05\,\mathrm{yr}$; broader parameter sweeps use $0.1\,\mathrm{yr}$, with selected thresholds rechecked at $0.05\,\mathrm{yr}$. These checks document the tested step-size sensitivity; they do not establish numerical stability or convergence for every parameter combination. Piecewise and threshold functions, including the positive-part operator $[x]_+$, are evaluated at each time step. The following state bounds are enforced during integration:

- $A_c \geq A_c^{\min}$ and $A_f \geq 0$;
- $q \in [0, q_{\max}]$;
- $P \geq 0$ and $D \geq 0$;
- $K \geq K_{\min}$.

Because the delay equations require histories over prior time intervals, $B(t)$ and $q(t)$ are initialized at their $t = 0$ values throughout the longest active delay window, $\max\{\tau_g, \tau_p\}$. In the controlled technology-wave experiment (SI §S8), the baseline and wave runs share the same pre-run history; the technology term changes current B, with its effect later entering the delayed carrying-capacity input $K(t - \tau_p)$. Floor-proximity values depend on the step size and numerical tolerance, so we report interval bands rather than artificially precise scalar thresholds.

All primary trajectories use delayed ecological regeneration, $G_c(q(t-\tau_g))$. SI §S3 compares the instantaneous-quality alternative $G_c(q(t))$; SI §S8 reports the matched technology-wave experiment and step-size checks.

3. Results

3.1. When conversion raises the total as capital land declines

For a given capital-land quality level q, its annual contribution per hectare, $\Upsilon_c(q)$, is a concave quadratic. It peaks at $q^* = q_{\max}/2$; with the baseline parameters, the maximum is $0.066\,\mathrm{gha\,ha^{-1}}$. The quality-related term

contributes 0.016 of that value, while direct yield contributes 0.05 gha ha^{-1}; the direct-yield component is larger in this case.

Conversion changes the aggregate according to the gap between the annual account contribution of one fast-land hectare and the effective value of one capital-land hectare. We call this gap the *composition diagnostic*:

$$\Delta b_{\text{conv}}(q, D) = b_f(D) - \Upsilon_c(q). \tag{15}$$

At baseline, Δb_{conv} equals +0.787, +0.784, and +0.794 gha ha^{-1} at $q = 0.3$, 0.6, and 0.9. Each lies in the composition-premium regime ($\Delta b_{\text{conv}} > 0$), where conversion adds to the aggregate at the margin. At $\Delta b_{\text{conv}} = 0$ the contribution is first-order neutral; below zero, it is a composition-discount. These labels describe the conversion term only. Maintenance, retirement, quality change, and other flows still determine the net change in B. At any given quality level, that contribution changes linearly with A_c; it has no interior maximum with respect to area.

Proposition 1 (Conversion, aggregate change, and what the total can reveal)**.** *Let $T_b \equiv 0$, $R_{rc} \equiv 0$ and hold D fixed, so that b_f is constant. Along any solution of (3)–(10),*

$$\dot{B} = \Delta b_{\text{conv}}(q)\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c}\, G_c'(q)\, A_c\, \dot{q}. \tag{16}$$

(i) During conversion ($u_c > 0$), A_c falls when restoration is off. The conversion component of $\dot{B}$ is $\Delta b_{\text{conv}}(q)u_c$, positive exactly when $\Delta b_{\text{conv}}(q) > 0$. It determines the sign of net $\dot{B}$ only if maintenance balances retirement ($\mu A_r = \eta_f A_f$) and quality is unchanged ($\dot{q} = 0$).

(ii) In the reduced comparison—with fixed demand and debt, baseline availability shares, balanced maintenance and retirement, and no quality change—a positive Δb_{conv} makes the shortfall shrink during conversion. The shortfall closes if the balance point is reached before the area floor binds. If the floor binds first, A_c approaches $A_c^{\min}$ and a shortfall may persist. Other active terms in the full model can change this result.

(iii) At fixed $b_f > 0$ and q, $B = b_f A_f + \Upsilon_c(q) A_c$ supplies one equation for the two land areas. Near an interior state, a small change in A_c can be offset by $\delta A_f = -(\Upsilon_c/b_f)\delta A_c$ and $\delta A_r = -\delta A_f - \delta A_c$, preserving both B and total area as long as the land bounds are respected. Thus several nearby allocations can have the same B; the aggregate alone does

not identify A_c. For a finite observed change ΔB, the offset is $\Delta A_f = (\Delta B - \Upsilon_c \Delta A_c)/b_f$. It is feasible only if all resulting areas remain within bounds, so the feasible range depends on the starting state. If A_f and b_f are observed independently, the residual $C = B - b_f A_f = \Upsilon_c(q) A_c$ identifies their product, not its two factors. An independent observation of A_c would give $\Upsilon_c(q) = C/A_c$. A single value of Υ_c cannot separate b_c, $b_{G,c}$, ρ_c, and q. Even with informative observations of q, it identifies at most b_c and the product $b_{G,c}\rho_c$, not $b_{G,c}$ and ρ_c separately.

Proof. Differentiate (3) with b_f constant and substitute (6)–(7) with $R_{rc} = 0$: $\dot{B} = b_f(u_c + \mu A_r - \eta_f A_f) - \Upsilon_c u_c + b_{G,c} G_c'(q) A_c \dot{q}$, which is (16). (i) Follows from (16) with $R_{rc} = 0$; when the maintenance–retirement imbalance and quality-growth terms vanish, only the conversion term remains. (ii) Under the reduced-case conditions in the statement, $S = E - B$ and $\dot{S} = -\dot{B} = -\Delta b_{\text{conv}} u_c < 0$ on $S > 0$. Since S is bounded below by zero and A_c is bounded below by $A_c^{\min}$, conversion either closes the shortfall or is limited by the floor. (iii) At fixed b_f and q, the observation map is linear; an interior feasible point admits the stated small perturbations along a level set of B, while finite changes must also obey the area bounds. The residual C identifies only $\Upsilon_c(q) A_c$ unless A_c is independently observed; the parameter combination follows from the definition of $\Upsilon_c(q)$. □

Figure 1a follows the baseline run with demand fixed at $E = 0.825$ gha and restoration off. Initial states are $A_f = 0.6$ ha, $A_c = 1.5$ ha, $A_r = 0.4$ ha, and $q = 0.9$. During conversion, model biocapacity B rises from 0.594 to 0.825 gha (39%); capital-land area falls from 1.500 to 0.808 ha (46%), while fast-land area grows from 0.600 to 0.923 ha.

Quality q rises from 0.9 to 1.0, moving farther above $q^* = 0.5$, where $\Upsilon_c(q)$ peaks. Its per-hectare value therefore falls from 0.05576 to 0.05000 gha ha^{-1}. The modeled capital-land contribution $\Upsilon_c(q) A_c$ falls from 0.08364 to 0.04038 gha (51.7%)—more than the 46.2% area decline because value per hectare also decreases. The one-to-one transfers conserve area: $\Delta A_f + \Delta A_c + \Delta A_r = 0$.

The adjustment is fastest early and then slows. On the deficit side, the approach to balance is exponential, with slow rate $|\lambda_1| = 0.054\,\text{yr}^{-1}$ and an e-folding time of about 19 yr. By year 25, 98% of the rise in B and 85% of the decline in A_c have occurred ($B = 0.820$ gha, $A_c = 0.913$ ha); by year 50, 97% of the area decline is complete. At the reported numerical resolution, $\dot{B} > 0$ and $\dot{A}_c < 0$ together for about 149 yr.

Reserve adjustment creates the long tail. As fast land retires to reserve, its contribution falls and a small shortfall reappears. Conversion then continues until $A_r/A_f = \eta_f/\mu$. The duration is not a fixed horizon: the slow rate λ_1 sets its timescale, while Δb_{conv} and the initial shortfall affect its size (SI §S2.2).

The aggregate has a clear identification limit: B alone cannot show whether A_c is shrinking while another component grows. Independent observations of fast-land area A_f and yield b_f would let us calculate $C = B - b_f A_f$. But C identifies only the capital-land contribution, not its area and per-hectare value separately.

We call a trajectory with rising B and falling A_c a *composition mask*. A related but distinct pattern is the *liquidation mask*, in which demand E stays fixed while A_c falls. The first compares aggregate biocapacity with land area; the second compares demand with land area. They can occur together, but neither implies the other. The diagnostic Δb_{conv} gives the sign of conversion's marginal contribution, not necessarily the net change in B when other terms are active; a liquidation mask does not require $\Delta b_{\text{conv}} > 0$. Observing B, E, and A_c distinguishes their defining patterns. Here, "liquidation mask" names a trajectory; it is not the optional timber-liquidation extension, which is off in every reported run.

3.2. Fixed demand leads to a minimum-area boundary, not a fold

Once population reaches carrying capacity, $P = K = B/e$ and demand $E = eP$ equals biocapacity B; the shortfall then closes. Conversion is therefore transient in the endogenous-population runs. To isolate how the balance point changes with demand, we also fix E and analyze a reduced land system. The constraint $A_c \geq A_c^{\min}$ is a *typed floor*: it protects one land class, not the aggregate.

Proposition 2 (Balance point, demand ceiling and affine deficit-side dynamics)**.** *For this reduced comparison, hold demand at E, switch off the technology boost and restoration, set debt $D = 0$, and fix quality at $q = q_{\max}$. This quality is the attractor of (10) for $q(0) > 0$. Let $k = \mu/(\mu + \eta_f)$ and*

$$B_{eq}(A_c) = b_{f0}\, k\, (A_{tot} - A_c) + b_c A_c = b_{f0} k A_{tot} - (b_{f0} k - b_c) A_c. \tag{17}$$

(i) With no conversion ($S = 0$), reserve-to-fast maintenance balances fast-land retirement. This gives $A_f = k(A_{tot} - A_c)$ and $A_r = (1-k)(A_{tot} - A_c)$, where $k = \mu/(\mu+\eta_f)$. The resulting total is $B_{eq}(A_c)$. If $b_{f0} k > b_c$,

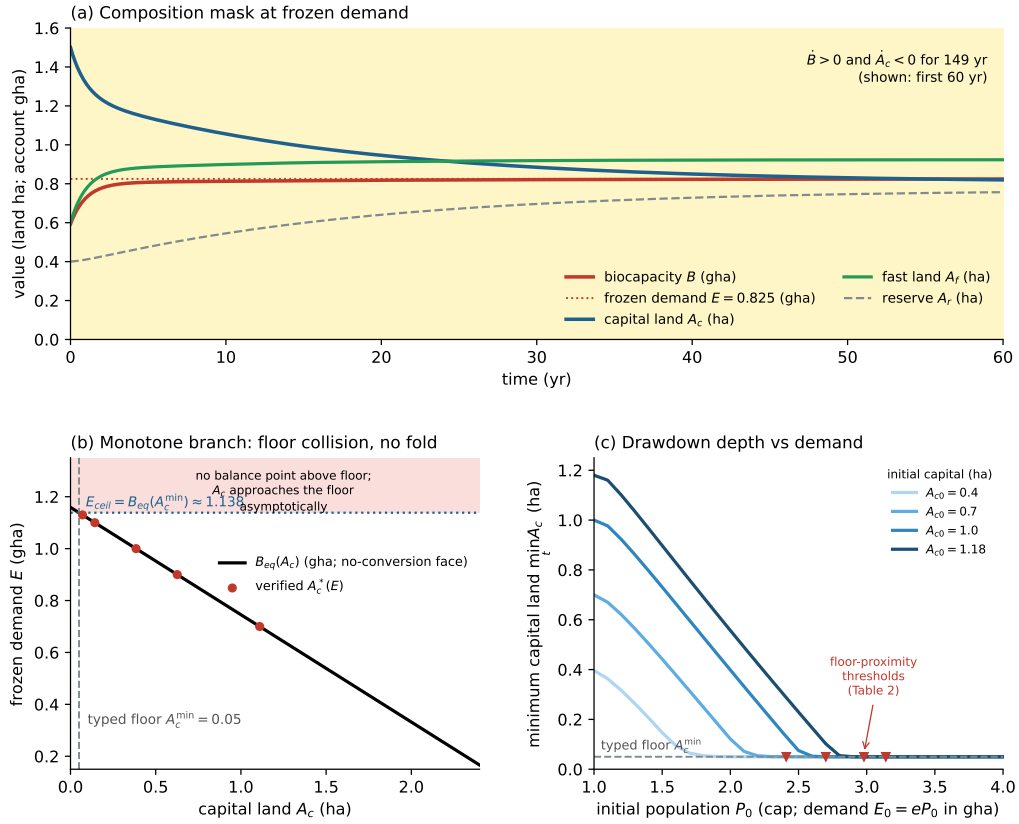


Figure 1: Baseline model results (Table 1). Account amounts B and E are in gha, land areas are in ha, and population is in cap. (a) Frozen demand ($E = 0.825$ gha), restoration off, and initial conditions $A_{f0} = 0.6$ ha, $A_{c0} = 1.5$ ha, and $q_0 = 0.9$. B rises from 0.594 to 0.825 gha as A_c falls from 1.500 to 0.808 ha and A_f rises from 0.600 to 0.923 ha. Shading marks the interval with $\dot{B} > 0$ and $\dot{A}_c < 0$ (about 149 yr; only the first 60 yr are shown). Within 25 yr, 98% of the rise in B and 85% of the fall in A_c occur. (b) Fixed-demand balance relation $B_{eq}(A_c)$; the listed balance areas are $A_c^*(E) = 1.110$, 0.626, 0.385, 0.143, and 0.070 ha at $E = 0.70$, 0.90, 1.00, 1.10, and 1.13 gha. The curve meets the minimum $A_c = 0.05$ ha at $E_{\text{ceil}} \approx 1.138$ gha; no balance point is admissible at higher demand. (c) Endogenous-demand runs with $\tau_g = 20$ yr, $\tau_p = 25$ yr, and $\chi_r = 0.10\,\text{gha}^{-1}\,\text{yr}^{-1}$. For each initial area A_{c0}, the minimum area reached varies continuously and monotonically with P_0 ($E_0 = eP_0$). The floor is approached asymptotically. Triangles mark the proximity values in Table 2, defined as within 10^{-6} ha of the floor; no discontinuity is observed.

this total decreases as A_c increases, so there is at most one balance point $B_{eq}(A_c^*) = E$:

$$A_c^*(E) = \frac{b_{f0} k A_{tot} - E}{b_{f0} k - b_c}. \tag{18}$$

The algebraic solution is feasible when $B_{eq}(A_{tot}) \le E \le E_{\text{ceil}}$, where $E_{\text{ceil}} = B_{eq}(A_c^{\min})$. It lies above the minimum area exactly when $E < E_{\text{ceil}}$. At lower demand, the algebraic root lies above A_{tot}, and every feasible no-conversion allocation already supplies at least E.

(ii) While shortfall is positive and conversion is not yet limited by the minimum-area constraint, the land subsystem is affine in $z = (A_f, A_c)$: $\dot{z} = Jz + c$, with

$$J = \begin{pmatrix} -\tau_{\text{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\text{conv}}} \\ \tau_{\text{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\text{conv}}} \end{pmatrix}, \qquad \det J = \frac{\mu + \eta_f}{\tau_{\text{conv}}} \Big(k - \frac{b_c}{b_f} \Big), \tag{19}$$

and $\operatorname{tr} J < 0$. The matrix does not depend on demand or on the current state. Its determinant is positive exactly when B_{eq} decreases, and its fixed point is the balance point in (i). For $B_{eq}(A_{tot}) \le E < E_{\text{ceil}}$, trajectories that remain in this regime approach that point at rates set by the eigenvalues of J. For $E > E_{\text{ceil}}$, the balance point would lie below the allowed minimum: A_c approaches $A_c^{\min}$ asymptotically and the shortfall tends to $S^* = E - E_{\text{ceil}} > 0$. Thus the reduced balance curve meets the minimum-area boundary; it does not turn back or form a fold. The leading eigenvalue does not approach zero at $E = E_{\text{ceil}}$, and the reduced equilibrium relation has no hysteresis. These conclusions hold under the stated assumptions and do not imply that the realized land allocation is independent of its history.

(iii) Above the demand ceiling, with degradation active, debt approaches a unique value $D^*(E)$ satisfying $\eta D = E - B_{eq}(A_c^{\min}; D)$, where $B_{eq}(\cdot; D)$ uses yield $b_{f0} e^{-\alpha D}$. This result requires $\eta > \alpha b_{f0} k (A_{tot} - A_c^{\min})$; the equilibrium debt increases with demand.

Proof. (i) With $S = 0$ and $R_{rc} = 0$, (7) gives $\mu A_r = \eta_f A_f$, hence $A_r / A_f = \eta_f / \mu$ and, with $A_r = A_{tot} - A_f - A_c$, $A_f = k(A_{tot} - A_c)$; substituting into (3)

with $q = q_{\max}$ ($\Upsilon_c = b_c$) and $D = 0$ gives $B = B_{eq}(A_c)$, and $S = 0$ requires $B \geq E$. Strict monotonicity and (18) follow from $b_{f0}k > b_c$. (ii) On the deficit side $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\text{conv}})$ is affine in z; substituting into (6)–(7) with $A_r = A_{tot} - A_f - A_c$ gives J and a constant vector c by direct differentiation; $\det J$ follows by expansion, and $\operatorname{tr} J = -\tau_{\text{conv}}^{-1}(1 - b_c/b_f) - \mu - \eta_f < 0$ for $b_c < b_f$. The fixed point solves $u_c = 0$ and $\mu A_r = \eta_f A_f$, i.e. it is the balance point. Because J is constant, convergence rates are constant along the branch. (iii) The map $D \mapsto \eta D + B_{eq}(A_c^{\min}; D) - E$ is continuous, negative at $D = 0$ when $E > E_{\text{ceil}}$, tends to $+\infty$, and is strictly increasing when $\eta > \alpha b_{f0} k (A_{tot} - A_c^{\min}) \geq -\partial_D B_{eq}$; hence it has a unique root, which increases with E. □

The uniqueness result concerns the reduced equilibrium relation; it does not make realized trajectories independent of their history. In the baseline, $R_{rc} = 0$ and conversion has no reverse flow. Once the shortfall closes, A_c stops falling and approaches its lower balance value. It remains below its pre-overshoot level because restoration is disabled. Meanwhile, q evolves separately, so $\Upsilon_c(q)$ can still change as A_c approaches that value. The reduced system therefore has a unique frozen-demand balance point under Proposition 2's assumptions, alongside a history-dependent post-overshoot allocation. This is not perpetual decline at equilibrium and is not a claim about real ecosystems.

At baseline, $k = 0.5455$, the balance-branch slope parameter is $b_{f0}k - b_c = 0.414\,\text{gha}\,\text{ha}^{-1}$, and the demand ceiling is $E_{\text{ceil}} = 1.138\,\text{gha}$. Equation (18) predicts balance areas of 1.110, 0.626, 0.385, 0.143, and 0.070 ha at demands of 0.70, 0.90, 1.00, 1.10, and 1.13 gha; these match the integrated equilibria (Fig. 1b). The eigenvalues of J are real and negative ($\lambda_1 = -0.0537$, $\lambda_2 = -0.998\,\text{yr}^{-1}$). Across the tested fast-land yields, λ_1 shifts only from -0.0522 at $b_f = 0.60$ to $-0.0556\,\text{yr}^{-1}$ at $b_f = 2.00$ and never changes sign (Table 2a).

Above the ceiling, the calculated debt equilibria are $D^* = 1.19$ and $5.38\,\text{gha}\,\text{yr}$ at $E = 1.158$ and $1.238\,\text{gha}$; numerical integration gives 1.21 and $5.39\,\text{gha}\,\text{yr}$. The uniqueness condition holds with margin: $\alpha b_{f0} k (A_{tot} - A_c^{\min}) = 0.034\,\text{yr}^{-1}$, below $\eta = 0.05\,\text{yr}^{-1}$. Under fixed demand, debt therefore saturates rather than growing without bound.

Critical slowing down refers to a fold approach in which the slowest return rate tends to zero and the return time $T_{ret} \approx -1/\lambda_1$ grows. That does not occur along this reduced balance branch: its linearization is constant, λ_1 stays away from zero, and the return time remains near 19 yr as E approaches E_{ceil}.

The modeled signal is instead A_c approaching its minimum—information an aggregate-only observation cannot provide.

When population responds to supply, the minimum A_c reached varies continuously and monotonically with initial population P_0 for each starting area A_{c0} (Fig. 1c). The trajectory approaches, but does not cross, the floor because $u_c \leq (A_c - A_c^{\min})/\tau_{\text{conv}}$. A larger initial capital-land area supports a larger P_0 before the run approaches that floor.

Table 2b gives the initial populations for which A_c comes within 10^{-2}, 10^{-4}, or 10^{-6} ha of its minimum. These tolerance-based values are proximity markers, not abrupt thresholds. At the tightest tolerance, $P_0 \approx 2.41$, 2.70, 2.98, and 3.14 cap for $A_{c0} = 0.40$, 0.70, 1.00, and 1.18 ha, respectively. They rise with starting area; no jump appears at any tested tolerance.

At $A_{c0} = 1.0$ ha, the 10^{-6} ha proximity value shifts by at most 0.001 cap across $\tau_g = 5$, 20, and 40 yr (SI Table S2). This small change applies only to the tested range; it does not show that delay is irrelevant more generally. The threshold is also unchanged to three decimal places when surplus-gated restoration is switched on (SI §S4), because that flow is inactive while $B < E$.

3.3. Technology gains are finite; debt depends on the shortfall

The technology boost and debt both change fast-land yield b_f, but in different ways. The boost has a fixed ceiling. Debt can accumulate while a shortfall persists, then approaches a finite level if that shortfall settles.

Proposition 3 (The technology boost is capped; debt depends on the shortfall)**.** *Under* (12)–(14)*:*

(i) The technology factor $(b_{f0} + T_b(t))/b_{f0}$ *stays between* 1 *and* $1 + \Delta b/b_{f0}$*. Its log-scale gain is therefore bounded by* $\ln(1 + \Delta b/b_{f0})$ *for all time (*0.78 *at baseline).*

(ii) Debt satisfies $D(t) \leq \max\{D(0),\ \sup_{s \leq t} [E - B]_+ (s)/\eta\}$*. If the shortfall settles at* S^**, then* D *tends to* S^*/η*. The resulting reduction in fast-land yield depends on the size of that sustained shortfall.*

(iii) The debt-related yield loss exceeds the largest possible technology gain once the sustained shortfall exceeds $S^\dagger = (\eta/\alpha)\ln(1 + \Delta b/b_{f0})$*, which is* 1.30 *gha at baseline.*

With additive degradation, $b_f = b_{f0} + T_b - \alpha D$*, the two effects enter symmetrically; the comparison in (iii) then reduces to* Δb *versus* αD^**. The asymmetry above follows from the model's multiplicative form.*

Table 2: Model results at the baseline parameter set. (a) At $q^* = q_{\max}/2$ ($\Upsilon_c(q^*) = 0.066\,\mathrm{gha\,ha^{-1}}$), the table reports $\Delta b_{\mathrm{conv}}(q^*)$, the eigenvalues of the deficit-side matrix J in (19), and the demand ceiling across a sweep of b_f; all other parameters are at baseline. (b) For endogenous-demand runs with $A_{f0} = 0.6\,\mathrm{ha}$, $q_0 = 0.9$, $\tau_g = 20\,\mathrm{yr}$, and $\tau_p = 25\,\mathrm{yr}$, the table reports the initial population P_0 at which the minimum A_c comes within the stated distance of the floor $A_c^{\min} = 0.05\,\mathrm{ha}$. The first two columns use a P_0 grid with step 0.1 ($\Delta t = 0.1\,\mathrm{yr}$, $T = 400\,\mathrm{yr}$); the final column uses bisection with the $10^{-6}\,\mathrm{ha}$ proximity criterion ($\Delta t = 0.05\,\mathrm{yr}$, $T = 1000\,\mathrm{yr}$). Because drawdown depth varies continuously with P_0 (Fig. 1c), these values define proximity bands, not a single threshold.

(a) Fast-land yield sweep (all rows composition-premium, $\Delta b_{\mathrm{conv}} > 0$)

b_f (gha ha^{-1})	$\Delta b_{\mathrm{conv}}(q^*)$ (gha ha^{-1})	λ_1 (yr^{-1})	λ_2 (yr^{-1})	E_{ceil} (gha)
0.60	+0.534	−0.0522	−0.975	0.804
0.85 (baseline)	+0.784	−0.0537	−0.998	1.138
1.20	+1.134	−0.0547	−1.014	1.606
2.00	+1.934	−0.0556	−1.029	2.675

(b) Initial population P_0 (cap) at which $\min_t A_c$ approaches $A_c^{\min}$ to within:

A_{c0} (ha)	within 10^{-2} ha	within 10^{-4} ha	within 10^{-6} ha
0.40	≈ 1.8	≈ 2.1	≈ 2.41
0.70	≈ 2.2	≈ 2.4	≈ 2.70
1.00	≈ 2.6	≈ 2.8	≈ 2.98
1.18	≈ 2.8	≈ 3.0	≈ 3.14

Proof. (i) is $0 \le T_b \le \Delta b$. (ii) From (13), $\dot{D} \le \bar{S} - \eta D$ with $\bar{S} = \sup_{s \le t} [E - B]_+$, so $D(t) \le \max\{D(0), \bar{S}/\eta\}$ by comparison; if $[E - B]_+ \to S^*$ then $D \to S^*/\eta$. (iii) compares $\alpha S^*/\eta$ with $\ln(1 + \Delta b/b_{f0})$. □

With fixed demand and a persistent shortfall, D approaches S^*/η; larger shortfalls imply higher debt. At baseline, the crossover $S^\dagger \approx 1.3\,\mathrm{gha}$ exceeds baseline B. Under this parameterization, the minimum-area constraint limits conversion before debt-related yield loss exceeds the largest possible technology gain. Debt nevertheless reduces fast-land yield over time.

A technology boost can raise B and carrying capacity; delayed population growth may raise demand too. Peak debt therefore depends on initial population and the summary window. In the matched SI §S8 experiment, for $P_0 = 1.0\,\mathrm{cap}$, peak debt over 0–250 yr is $0.02020\,\mathrm{gha\,yr}$ with the boost and $0.00813\,\mathrm{gha\,yr}$ without it (ratio 2.49). For $P_0 = 1.2\,\mathrm{cap}$, the full-run maximum is lower with the boost (0.05104 versus $0.06122\,\mathrm{gha\,yr}$), but the maximum after the 40 yr wave midpoint is higher (0.02518 versus $0.01085\,\mathrm{gha\,yr}$; ratio 2.32). These are conditional model results, not forecasts or empirical estimates.

3.4. Recovery requires an explicit, gated restoration flow

Conversion moves land from capital to fast use. The model has no route back to capital land unless restoration R_{rc} is active; fast-land retirement returns land to reserve instead. Lowering population, footprint, or shortfall therefore does not, by itself, reverse earlier conversion. With $\chi_r = 0$, A_c stays at its post-conversion level.

In one endogenous-demand run ($A_{c0} = 1.0\,\mathrm{ha}$, $P_0 = 2.0\,\mathrm{cap}$), A_c drops to 0.398 ha within four years and remains there without restoration. With surplus-gated restoration ($\chi_r = 0.10\,\mathrm{gha^{-1}\,yr^{-1}}$), it recovers to 0.828 ha, a recovery fraction of $\mathcal{R}_c = (A_c(T) - A_c^{\mathrm{deg}})/(A_c^{\mathrm{init}} - A_c^{\mathrm{deg}}) = 0.71$ (SI §S4.3).

The two restoration gates operate under different conditions. Surplus-gated restoration is inactive during a shortfall, but may continue after a surplus opens while population adjusts. Reaching a specified endpoint therefore requires a target-limited rule (SI §S4.4). Deficit-gated restoration acts during shortfall, when supply is already insufficient. At the tested rates, conversion outpaces restoration and drawdown continues. The result depends on timing and rates; neither gate is universally preferable.

3.5. NFA account composition, 1961–2023

For the primary empirical analysis, we use complete world observations in NFA Edition 2026, results and metadata version 1.0, and exclude the estimated 2024–2025 values. Our 1961–2022 sensitivity uses the same account edition and population series.

Between 1961 and 2023, total biocapacity account value B rises from 9.7254×10^9 to 12.1428×10^9 gha (24.86%; $\Delta \ln B = +0.221997$). Population grows from 3.06487 to 8.09173 billion. Over the same interval, B/P falls from 3.17319 to 1.50065 gha person^{-1} (52.71%; $\Delta \ln(B/P) = -0.748840$). The Global Ecological Footprint of Production per person (E_{prod}/P) rises from 2.19700 to 2.56462 gha person^{-1} (16.73%). Meanwhile, E_{prod}/B rises from 0.69236 to 1.70901, first exceeding one in 1973.

The exact identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$ yields $+0.970837 - 0.748840 = +0.221997$. It separates population-scale change from change in biocapacity per person, but does not explain their causes. The same-edition 1961–2022 sensitivity gives $\Delta \ln B = +0.219497$, a 52.41% fall in B/P, a 17.70% rise in E_{prod}/P, and 1973 as the first year with $E_{\text{prod}}/B > 1$ (SI §S5).

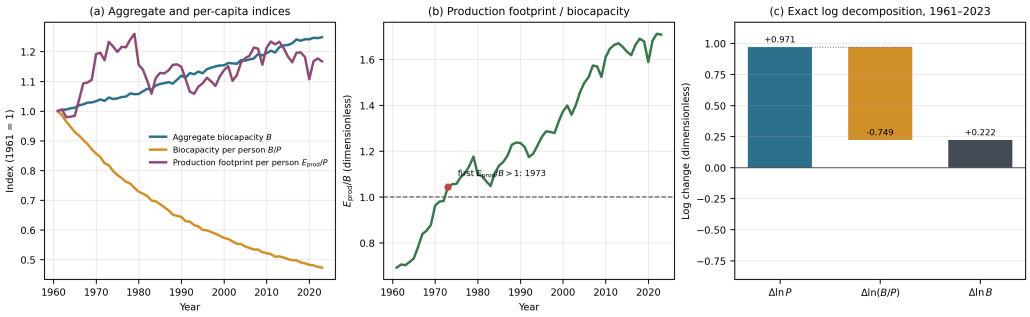


Figure 2: Aggregate and per-capita movements in NFA Edition 2026 (world, 1961–2023; complete observations). B is total biocapacity, E_{prod} is the workbook's Global Ecological Footprint of Production, and P is the single UN PopStat population series; per-capita values are calculated from totals. (a) Indices (1961 = 1) of B, B/P, and E_{prod}/P. (b) E_{prod}/B, first above one in 1973. (c) Exact log identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$. Account values are in gha; plotted indices, ratios, and log changes are dimensionless. The 2024–2025 estimated rows are excluded.

The component series show how NFA account composition changed (Fig. 3; Table 3). Cropland contributes +24.06 percentage points to the aggregate's +24.86% change—about 96.8% of the net percentage change. Forest biocapacity contributes −5.95 points, grazing land −0.27, fishing

grounds −0.42, and built-up land +7.44. The non-cropland account sum, which includes built-up land, is nearly unchanged (1.0092×, or +0.80 points) because the built-up increase offsets declines elsewhere. These are changes in account values. They do not establish changes in forest biomass, soil condition, wetland extent, fish biomass, or model area A_c. Built-up land is an account class, not ecological capital. Component values sum to total biocapacity within workbook rounding at both endpoints.

Table 3: NFA Edition 2026 biocapacity-account endpoint ratios and contributions. Contributions are $100\,[B_i(T) - B_i(1961)]/B(1961)$ percentage points; endpoint ratios are $B_i(T)/B_i(1961)$. The 2022 sensitivity is from the same edition. All entries are account values, not physical-stock measurements.

Component	2023 ratio	2023 contribution (pp of B_{1961})	2022 sensitivity ratio	2022 contribution (pp of B_{1961})
Cropland	2.8701	+24.0577	2.8514	+23.8183
Grazing land	0.9828	−0.2688	0.9815	−0.2890
Forest biocapacity	0.8996	−5.9525	0.9019	−5.8137
Fishing grounds	0.9627	−0.4212	0.9596	−0.4559
Built-up land†	8.7935	+7.4416	8.6298	+7.2854
Non-cropland sum†	1.0092	+0.7990	1.0083	+0.7267
Total biocapacity	1.2486	+24.8568	1.2454	+24.5450

†Built-up land is not ecological capital; the non-cropland account sum includes it. Component sums match total biocapacity within workbook rounding.

We also use external cropland area and cereal yield to build an illustrative proxy for the fast-land contribution (SI §S5). Anchoring it to the measured 1961 NFA cropland share ($\phi_0 = 0.128647$, about 12.86%), the exact log-change split assigns +0.268662 to the cereal-yield index, +0.037710 to cropland area, and −0.084375 to the residual. Together these sum to $\Delta \ln B = +0.221997$. The identity is exact for this proxy, but its allocation changes with the anchor and is not a physical-stock attribution.

3.6. Falsifiable predictions

The model yields five testable predictions. Testing its composition mechanism requires more than aggregate biocapacity: it requires information on capital-land area, fast-land area, and relevant land-cover or yield proxies.

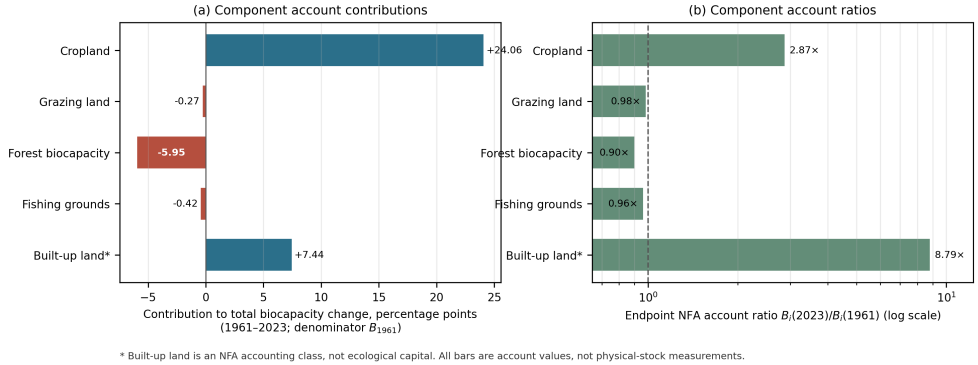


Figure 3: NFA Edition 2026 land-type biocapacity-account changes, world, 1961–2023. (a) Component contributions in percentage points of B_{1961}. (b) Endpoint ratios $B_i(2023)/B_i(1961)$. "Forest biocapacity" is the workbook's exact label. All bars are account values, not measurements of physical stocks; built-up land is an NFA accounting class, not ecological capital.

P1. **Conversion can raise the total under a specific condition.** In the baseline frozen-demand run, $\dot{A}_c < 0$ while $\dot{B} > 0$ for a finite interval (Fig. 1a). Conversion adds to B exactly when $\Delta b_{\text{conv}} > 0$, although other flows, quality, debt, technology, and restoration determine the net change. With baseline shares $\sigma_f = \sigma_c = 1$, conversion requires demand to exceed total biocapacity ($R_B = E/B > 1$). The reserved-share extension can permit conversion when $R_B \leq 1$, but that is not a baseline result. A composition mask is rising B with falling A_c; a liquidation mask is steady E with falling A_c. The two patterns may overlap but are not interchangeable. The sign of Δb_{conv} describes conversion's contribution, not necessarily the net change in B.

P2. **The duration depends on model rates, not a universal horizon.** The slow return rate λ_1 (along with τ_{conv}, μ, η_f, and b_c/b_f) sets the timescale; Δb_{conv} and the initial shortfall affect the size of the change. In the baseline, most conversion occurs within one to two e-folding times, when the remaining gap shrinks by a factor of e.

P3. **The reduced model has no fold-type early warning.** Return time does not grow as demand approaches the ceiling, and drawdown depth changes continuously with demand (Fig. 1c). The model instead signals that A_c is approaching its minimum. It predicts no fold-related warning

before that boundary.

P4. **The population proximity value depends on starting area and is weakly delay-sensitive in the tested range.** The initial population at which a run approaches the capital-land floor rises with starting A_c (Table 2b). For $A_{c0} = 1.0\,\text{ha}$, the $10^{-6}\,\text{ha}$ proximity value shifts by at most $0.001\,\text{cap}$ across $\tau_g = 5$, 20, and $40\,\text{yr}$ (SI Table S2). This comparison does not establish delay invariance beyond that range; the small shift is consistent with the modest regenerative contribution and monotone quality path in the baseline.

P5. **Recovery requires a return flow.** Lower demand alone does not restore capital-land area. The model needs an explicit restoration flow. A surplus-gated flow is inactive during shortfall, whereas a deficit-gated flow can further reduce provision.

Aggregate B alone cannot test these predictions. The composition mechanism requires independent observations or defensible proxies for A_c and fast-land yield b_f. Testing delay, recovery, and demand-threshold predictions also requires relevant time series and intervention or demographic data.

4. Discussion

4.1. What the two-land model shows

The two-land model yields three results relevant to human–environment models (Schaefer, 1954; May, 1973; Meadows et al., 1972, 2004; Wackernagel and Rees, 1996; Brander and Taylor, 1998). First, conversion can add to total biocapacity while A_c falls. Whether B rises overall also depends on other land flows and quality change; Proposition 1 gives the exact condition for the conversion term. The technology boost is capped, while debt depends on the magnitude and duration of shortfall (Proposition 3).

Second, under the reduced fixed-demand assumptions ($q = q_{\max}$, $D = 0$, technology boost off, and no restoration), the balance relation declines steadily with A_c until it meets the area floor. It does not fold back. The absence of a fold, hysteresis, or critical slowing applies to this equilibrium relation, not to every trajectory. With one-way conversion and no return flow, A_c remains below its pre-overshoot level after conversion stops, although it no longer declines at the balance point (Table 4). Third, recovery requires an explicit return flow; its effect depends on whether the restoration gate opens during a shortfall or a surplus.

SI §S7 gives a separate one-stock comparator. Its state A has comparator-specific units; it is not A_c. With constant parameters and no debt, its equilibrium relation is $B(A) = (b + b_G\rho)A - (b_G\rho/A_{\max})A^2$, with $P = B(A)/e$. At the illustrated settings, $b_G\rho < b$, so the relation is monotone; an interior maximum would require other parameters. Its neutral equilibrium family and selected delay trajectories belong only to that formulation. They establish neither a fold nor a universal critical delay or collapse mechanism in the two-land model.

These results also clarify when to expect common early-warning indicators. Rising autocorrelation and variance are associated with approaching a fold (Scheffer et al., 2001). This reduced model reaches a minimum-area boundary instead, so those indicators need not rise first. Their absence would not establish safety. The relevant model signal is the path of A_c, which the aggregate alone cannot reveal.

Table 4: Equilibrium, trajectory, and tested delay-sensitivity properties under the stated assumptions.

Property	Finding	Scope
Fold on the reduced frozen-demand branch	Absent	$B_{eq}(A_c)$ is strictly monotone; the balance point collides with the typed floor at $E_{\text{ceil}} \approx 1.138\,\text{gha}$ (Proposition 2)
Critical slowing as $E \to E_{\text{ceil}}$	Absent	The slow eigenvalue remains negative and constant along the reduced branch
Hysteresis of the frozen-demand equilibrium relation	Absent	There is at most one feasible balance point; this does not imply history independence of realized states
Path dependence of realized A_c	Present, conditional	With $R_{rc} = 0$, one-way conversion leaves A_c below its pre-overshoot level after shortfall closure, while $\dot{A}_c = 0$ at equilibrium
Discontinuous response to endogenous demand	Not observed	In the reported runs, minimum A_c varies continuously and monotonically with initial population; proximity thresholds depend on tolerance
Regeneration-delay sensitivity (τ_g, with $\tau_p = 25\,\text{yr}$)	Small in tested range	At $A_{c0} = 1.0\,\text{ha}$, the delayed-quality 10^{-6} ha threshold varies by 0.001 cap across $\tau_g = 5, 20, 40\,\text{yr}$ (SI Table S2); this does not establish invariance outside that range

4.2. Measurement implications

Biocapacity in gha is a value-weighted total, not an inventory of land classes. A 20% increase cannot distinguish modest gains across all components from a large gain in one that offsets losses elsewhere. Proposition 1(iii) formalizes the limit: even with fixed per-hectare yields, the total and the conserved-area constraint leave multiple feasible allocations among fast, capital, and reserve land.

A related transition-assessment study defines a linear compensatory operator and compares perfect substitutability with less-substitutable CES alternatives (Sections 2 and 4 of Abaee, 2026b). ECOMOD's B is also a linear sum, but the companion analyzes a different, finite-plan setting. There, the compensatory protocol accepts states even when no single available plan meets every component floor (Theorem 2 in Abaee (2026b)). Theorem 4(iii) in Abaee (2026b) identifies an exact interval of diagonal-gap states on its witness-defined region that each positive elasticity falsely certifies as safe; thus no positive-elasticity member is uniformly safe on that region. If feasibility is required throughout a transition, falling below a land-class minimum remains a violation even if the endpoint later recovers.

For the NFA, these findings concern account composition, not physical stocks. In Edition 2026, total world biocapacity account value B rises 24.86% from 1961 to 2023, while B/P falls 52.71%. The Global Ecological Footprint of Production per person (E_{prod}/P) rises 16.73%, and E_{prod}/B first exceeds one in 1973. B and E_{prod} are annual account amounts in gha, with the annual basis implicit; per-person values are in gha per person, and E_{prod}/B is dimensionless.

Cropland contributes +24.06 percentage points to the total change; Forest biocapacity contributes −5.95, grazing land −0.27, fishing grounds −0.42, and built-up land +7.44. The non-cropland sum is nearly flat because these account movements offset one another, including the large built-up-land increase. SI §S5 reports the same-edition 1961–2022 comparison. None of these values measures forest biomass or condition, soil carbon, wetland extent, fish biomass, or another physical stock. In particular, a decline in Forest biocapacity does not establish a decline in forest biomass. Built-up land is an NFA accounting class, not ecological capital. The ratios, contributions, vintage, and endpoint sensitivity describe account composition only (Lo et al., 2026; Borucke et al., 2013; Blomqvist et al., 2013; Giampietro and Saltelli, 2014; Galli et al., 2016; van den Bergh and Grazi, 2015).

The model result is distinct from the NFA trends. In the frozen-demand baseline (Fig. 1a), A_c falls from 1.500 to 0.807659 ha (46.16%), while model B rises from 0.594 to 0.825 gha (39%). Quality q rises from 0.9 to 1.0, moving farther above $q^* = q_{\max}/2$, where $\Upsilon_c(q)$ peaks. Its per-hectare value therefore falls from 0.05576 to 0.05000 gha ha^{-1} (10.33%), and the modeled contribution $\Upsilon_c(q)A_c$ falls from 0.08364 to 0.04038 gha (51.72%).

With restoration off, one-way conversion leaves A_c below its starting value after the shortfall closes. It then remains at the lower balance level rather than declining indefinitely. This is a conditional model result, not an observation or estimate of global physical-stock change.

We assembled no common-boundary global physical-stock series for these components and therefore report no physical-stock composite or global decline finding. The SI scoreboard is a specification—not an inventory—with candidate stock, extent, condition, and annual-flow measures and the requirements for comparing them; it contains no endpoint values. Until spatial and temporal boundaries, units, uncertainty, overlap, valuation or normalization, and permitted substitution are specified, these unlike measures should remain a nonadditive vector. SEEA Ecosystem Accounting also reports ecosystem extent and condition separately from ecosystem services and monetary presentations (United Nations et al., 2021).

The external cropland-area and cereal-yield series provide an illustrative proxy, not direct measurements of the model's land classes. The proxy is anchored to the 1961 NFA cropland share; cereal yield is a productivity sentinel, not an all-crops measure, and the area and account boundaries do not match by construction. External cropland area A_f is in hectares and index-scaled yield $\hat{b}_f$ in gha ha^{-1}, so $X = \hat{b}_f A_f$ and $C = B - X$ are account amounts in gha. The residual can absorb boundary and calibration differences, other account components, and conversion-factor changes. It is neither $\Upsilon_c(q)A_c$ nor a physical-stock measure. Physical interpretation would require independent, boundary-consistent measures of area, condition, yield, and regeneration, together with a validated mapping and valuation. Forest area could inform one part of the effective capital class, but cannot represent the entire class or its condition.

4.3. Policy controls and limits

We compare three controls. A conversion stop triggered at $A_c = A_c^{\min}$ preserves the capital-land floor. A ceiling on fast-land area A_f does not guarantee that floor unless the stop rule is explicitly tied to it and to the

remaining-area constraints. Nor can a heuristic governor that monitors only total B guarantee the floor: the aggregate does not identify A_c.

The restoration gates also behave differently. Surplus-gated restoration is inactive during a shortfall; deficit-gated restoration acts during the shortfall and can reduce provision further. This comparison echoes the distinction between protective and mobilizing institutional feedback in related work (Abaee, 2026a). With a common reserved share, $\sigma_f = \sigma_c = \sigma < 1$, conversion is inactive while $E \leq \sigma B$. With separate shares, the condition is $E \leq \sigma_f Y_f + \sigma_c Y_c$. This reservation applies to annual human-available flow, not to spatially protected area. Half-Earth, by contrast, is a spatial conservation target (Wilson, 2016). The connection is only the abstract idea of reservation: the model neither maps protected areas nor evaluates that proposal. Robustness to disturbances and parameter uncertainty remains untested.

4.4. Relation to prior work

The ledger companion distinguishes weak and strong sustainability as operating regimes according to whether material loops close at throughput. In its terms, a positive weighted sum ranks outcomes under weak sustainability, whereas componentwise adequacy is the strong-sustainability predicate (Abaee, 2026c, Sections 1.1 and 10.1). ECOMOD's B is an annual biocapacity-account total. The link is an analogy about compensation, not an identity with the companion's material-flow ledger or a complete sustainability classification. The assessment companion studies aggregate measures and typed transition criteria in a separate finite-plan setting (Abaee, 2026b); the governance companion studies institutional feedback (Abaee, 2026a). The delays here represent ecological regeneration and demographic response, not institutional review.

4.5. Limitations

Several limitations set the scope of the results. Ecological-capital land combines unlike assets, so its single quality state, regeneration rule, composition diagnostic, and demand ceiling are effective model quantities—not estimates for forests, soils, or wetlands. Most parameters are representative rather than fitted. A cross-section cannot identify b_c, $b_{G,c}$, and ρ_c separately. Recovery records from different systems provide timescale context, not a global estimate of ρ_c; they describe regeneration responses, not the delay τ_g. We hold the per-person footprint constant to isolate feedbacks among land,

biocapacity, and demand. A changing footprint would require another model extension.

The composition pattern depends on demand and on the mechanisms included here; under the stated assumptions, it is bounded. The exact shortfall switch makes stability at $S = 0$ one-sided. Drawdown thresholds are reported as proximity bands because they depend on tolerance and time step. We fix debt repayment at $\eta = 0.05\,\mathrm{yr}^{-1}$; the zero-repayment limit ($\eta \to 0$), which removes the debt equilibrium, is not analyzed. NFA accounts combine area and yield and therefore cannot identify them separately. Finally, replacing delayed regeneration $G_c(q(t-\tau_g))$ with current-quality regeneration $G_c(q(t))$ lengthens the demonstration mask by 1.3 yr and shifts the initial-population threshold by at most 0.001 cap (SI §S3).

5. Conclusions

Aggregate biocapacity cannot reveal how its underlying land classes are changing. In the two-land model, conversion raises total B at the margin when fast-land contribution per hectare exceeds capital land's effective value ($b_f > \Upsilon_c(q)$), while A_c falls if restoration is off (Proposition 1). Other model terms determine whether total B rises overall. This is a conditional accounting mechanism, not an empirical census of global physical stocks.

The reduced fixed-demand comparison follows a single balance branch to the specified minimum-area boundary without a fold, hysteresis, or critical slowing. That result is limited to the stated comparison and does not characterize every trajectory of the delayed model. Realized paths remain history-dependent: when shortfalls subside and restoration is off, A_c remains below its pre-overshoot level, but it does not continue to decline at the balance point. In the representative baseline run, total model biocapacity rises 39% while ecological-capital area falls 46.16%, its per-hectare quality term $\Upsilon_c(q)$ falls 10.33%, and its modeled contribution $\Upsilon_c(q)A_c$ falls 51.72%.

Separately, NFA Edition 2026 world account values rise 24.86% between 1961 and 2023, while biocapacity per person B/P falls 52.71% and the production footprint per person rises 16.73%. Cropland contributes +24.06 percentage points, Forest biocapacity −5.95, and Built-up land +7.44 as an accounting class. These normalized annual account values are not observations of physical stocks or model A_c.

No common-boundary global physical-stock series was assembled, so we report no composite or global physical-stock decline finding. The SI scoreboard sets out candidate measures and comparison requirements; it is not an empirical inventory. Physical measures with different units and boundaries cannot be combined into one stock total without explicit treatment of uncertainty, overlap, valuation or normalization, and allowed substitution. The auxiliary proxy residual $C = B - \hat{b}_f A_f$ is neither a physical-capital measure nor the model contribution $\Upsilon_c(q)A_c$; a future comparison would require independent observations and a validated mapping.

The practical lesson is to report component accounts alongside the aggregate. Returning to the orchard metaphor, a ledger of annual harvest alone cannot tell whether the trees, roots, and soil that support future harvests have changed. That physical record requires independent, comparable observations; it cannot be inferred from the account total. A harvest total is not a census of the orchard.

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Declarations

Data availability. All empirical inputs are publicly available from the sources cited in the manuscript. Derived analysis data and model code will be deposited in a public GitHub repository and archived in Zenodo; repository and record details will be added after the deposit is complete.

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